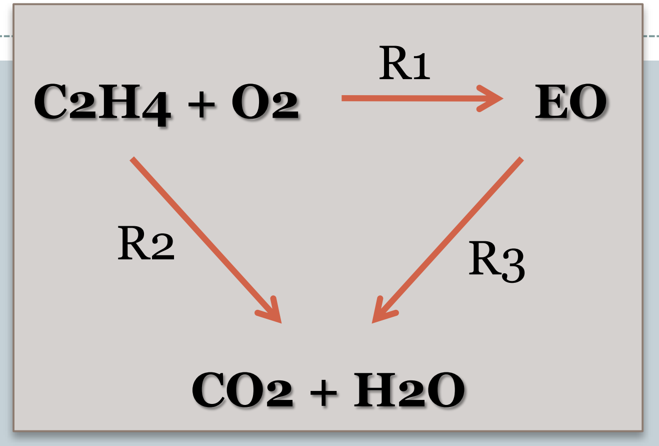
CH2 – CH2
O

Ethylene Oxide



$$2CH_2{=}CH_2 + O_2 \xrightarrow[\substack{250-300^oC \\ 30\ bar}]{Ag} 2\ CH_2 - CH_2\ (O)$$

$\triangle$ H = - 35.2 Kcal

Competing Reaction

$$CH_2 = CH_2 + 3O_2 \longrightarrow 2CO_2 + 2H_2O$$

$\triangle$ H = - 339.6 Kcal

- **Selectivity is the ratio of moles of ethylene oxide produced per mole of ethylene reacted.**
- **Heat of reaction increases with decrease in selectivity . Therefore, the heat removal system must be sufficiently adaptable to cope with growing heat release in the reactor at rising reaction temperatures during the aging process of the catalyst, which is characterized by a decrease in its selectivity**

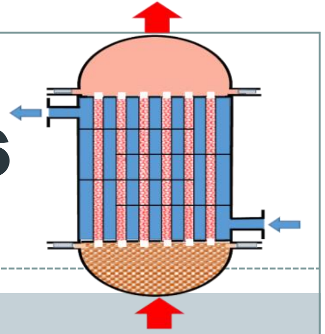
Ethylene oxide is a colorless and flammable gas with a faintly sweet odor.

End Uses

Approximately 50 % of sterile medical devices are treated with EO. EO is also used to sterilize some food products such as spices, certain dried herbs, dried vegetables, sesame seeds and walnuts.

- **Ethylene glycol** **HO-CH2-CH2-OH**
- **Detergents**
- **Ethanol amines** **HO-CH2-CH2-NH2 (mono)**
- **Glycol ethers**
- **Poly glycols**
- As a poison gas that leaves no residue on items it contacts, pure ethylene oxide is a **disinfectant** that is widely used in hospitals and the medical equipment industry to replace steam in the sterilization of heat-sensitive tools and equipment, such as disposable plastic syringes

Typical Reaction Conditions


Catalytic Fixed Bed (TFXB)

- Temp.: 200 - 300 °C
- Catalyst: Ag_2O
- Residence time: 1 sec
- Selectivity: (oxygen-based) 70 – 75 mole % ethylene
  (air-based) 62 – 72 mole % ethylene


Hollow cylinders of Ag_2O Catalyst

Quantitative Requirements

Basis: 1 ton ethylene oxide (99% purity)

• Ethylene	0.92 ton
• Air	9.0 ton
• Silver	0.3 Kg
• Ethylene dichloride (side reaction suppressor)	10 – 15 Kg
• Steam	0.1 ton
• Water	180 ton

Process Description

- **Ethylene and air are compressed separately, mixed together and passed over a <u>catalyst of silver oxide</u> in a <u>fixed-bed tubular reactor</u>.**

- **The main reactor consists of thousands of catalyst tubes in bundles. These tubes are generally 6 to 15 m long with an inner diameter of 20 to 50 mm. The catalyst packed in these tubes is in the form of spheres or rings of diameter 3 to 10 mm. The operating conditions of 200°C to 300°C with a pressure of 1 to 3 MPa prevail in the reactor.**

- **To maintain this temperature, the cooling system of the reactor plays a vital role. With the aging of the catalyst, its selectivity decreases and it produces more exothermic side products of CO_2.**

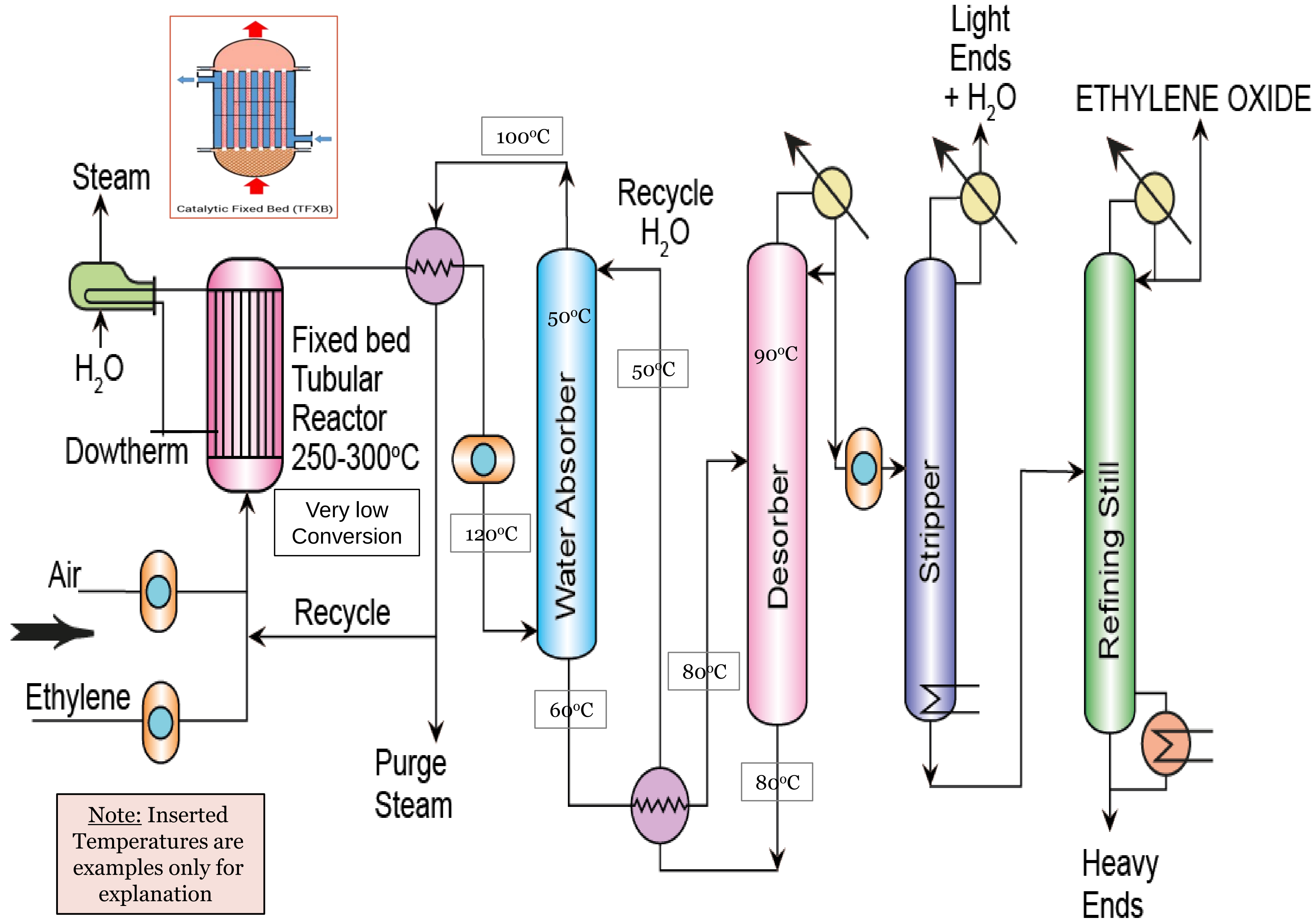
Process Description (cont.)

- The reaction is exothermic, so it is best carried in a tubular reactor where salt or dowtherm (heat transfer fluid) is pumped inside the shell around the tubes to maintain the temp. at 250-300^{o}C.
- A side-reaction suppressing agent (ethylene dichloride) is added to the feed to reduce competitive oxidation reaction to CO_2+H_2O.
- Heat is recovered in a WHSB where steam is produced and the cooled dowtherm is recycled back to the shell of the reactor .
- Effluent gases from the main reactor containing ethylene oxide (1–2%) and CO_2 (5%) along with some amounts of CO, N_2, CH_2=CH_2, CH_4, are fed to a water absorber (ethylene oxide scrubber) under pressure where majority of ethylene oxide is absorbed then sent to a desorber fractionator tower and taken overhead.

Process Description (cont.)

- **This stream is compressed, and then sent to a fractionating tower (stripper) where light ends and water are removed at the top.**

- **The bottom stream (ethylene oxide + heavy ends) are sent to another fractionating tower where heavy ends are separated at the bottom and ethylene oxide is produced at the top of the refining still.**



Ethylene Oxide by air oxidation of ethylene

Major Engineering Problems

➢ Volume Ratio of Air-Ethylene

- Lower explosion / flammability limit (LEL / LFL) for ethylene oxide in air = 3%

To overcome

- Recycle inerts from the absorber in order to keep concentration of C_2H_4 at or below this value.
- Use N_2 carrier stream.
- **Note:** Side stream purge removes some CO_2 and H_2O.

Major Engineering Problems (cont.)

➢ Use of Reactors in Series

- The ethylene conversion is very low so two reactors may be used in series to allow higher ethylene conversion. This takes place only after an economic balance between cost of ethylene and fixed charges of the additional reactor.

Major Engineering Problems (cont.)

➤ Air Versus Oxygen

- On using O_2 , space-time yield for a given reactor volume is increased 3-4 times.
- Also the absorber can be small because product concentration is 6 times greater than with air.
- Oxygen is only used when plants are located near a low-cost O_2 source.
- Most plants are based on air-oxidation.

Major Engineering Problems (cont.)

➢ **Fluidized-Bed versus Fixed-Bed**

- Fluidized bed allows perfect heat control.

However:

- A fluidized bed needs high residence time for mixing.
- Catalyst sintering may clog tubes.
- Ag_2O may be transferred to the walls of the reactor.

Solution:

Use Fixed bed tubular reactor

Major Engineering Problems (cont.)

➢ <u>Hydration of ethylene oxide to glycol</u>

- $CH2 - CH2 + H2O \xrightarrow{150^{o}C} HO - CH_2 - CH_2 - OH$

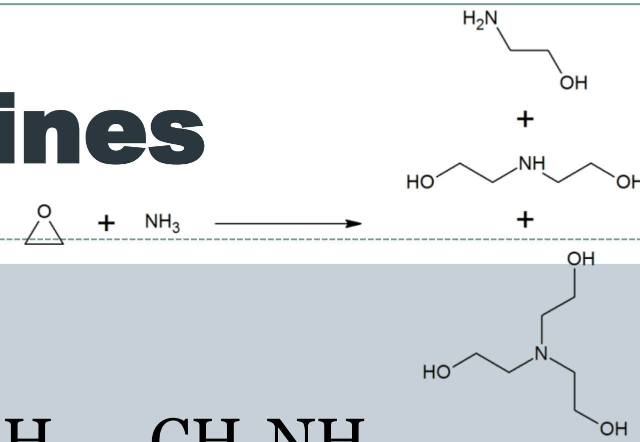
  O

  Ethylene oxide Ethylene glycol

- **Although water is used in the absorbing-desorbing train, this reaction almost does not take place as hydration rate constant is insignificant at <u>low temperature</u>.**

Reaction with NH3 is called Ammonolysis which is an exothermic reaction

Ethanol Amines



$$\underset{\text{O}}{CH_2 - CH_2} + NH_3\ \boxed{(H.NH_2)} \longrightarrow HOCH_2 - CH_2NH_2$$

Ethylene oxide

mono ethanol amine (MEA)

$$\boxed{(HOCH_2 - CH_2)\ H.NH}$$

$$\underset{\text{O}}{CH_2 - CH_2} + \underset{\text{MEA}}{HOCH_2 - CH_2NH_2} \longrightarrow (HOCH_2 - CH_2)_2\ NH$$

diethanol amine (DEA)

$$\boxed{(HOCH_2 - CH_2)2\ H.N}$$

$$\underset{\text{O}}{CH_2 - CH_2} + \underset{\text{DEA}}{(HOCH_2 - CH_2)2NH} \longrightarrow (HOCH_2 - CH_2)_3\ N$$

triethanol amine (TEA)

End Uses

- **<u>Sweetening of acid gases</u>**
- **Production of detergents**
- **Corrosion inhibitors**
- **Foam stabilizers**
- **Automobile and metal polishes**
- **Floor and wood work cleaner**
- **Hand sanitizer**
- **Dish washer liquid**
- **Shaving cream**

Kinetics of Complex Series Reactions

$$\frac{dC_M}{dt} = k_1 C_E C_A - k_2 C_E C_M$$

$$\frac{dC_D}{dt} = k_2 C_E C_M - k_3 C_E C_D$$

$$\frac{dC_T}{dt} = k_3 C_E C_D$$

MEA	monoethanolamine
DEA	diethanolamine
TEA	triethanolamine

E: ethylene oxide
A: ammonia
M: MEA **D: DEA** **T: TEA**

Kinetics of Complex Series Reactions (cont.)

Weight Ratios of Ethanol amines	Moles of Ammonia : Ethylene Oxide		
	10 (10:1)	**0.5** (1:2)	**0.1** (1:10)
MEA	**75 – 61 %**	**25 – 31 %**	**12 – 15%**
DEA	**21 – 27 %**	**28 – 32 %**	**23 – 26 %**
TEA	**4 – 12 %**	**47 – 37 %**	**65 - 59 %**

Kinetics of Complex Series Reactions (cont.)

- **Specific rate** constants depend on **temperature** and reactant **concentration**.

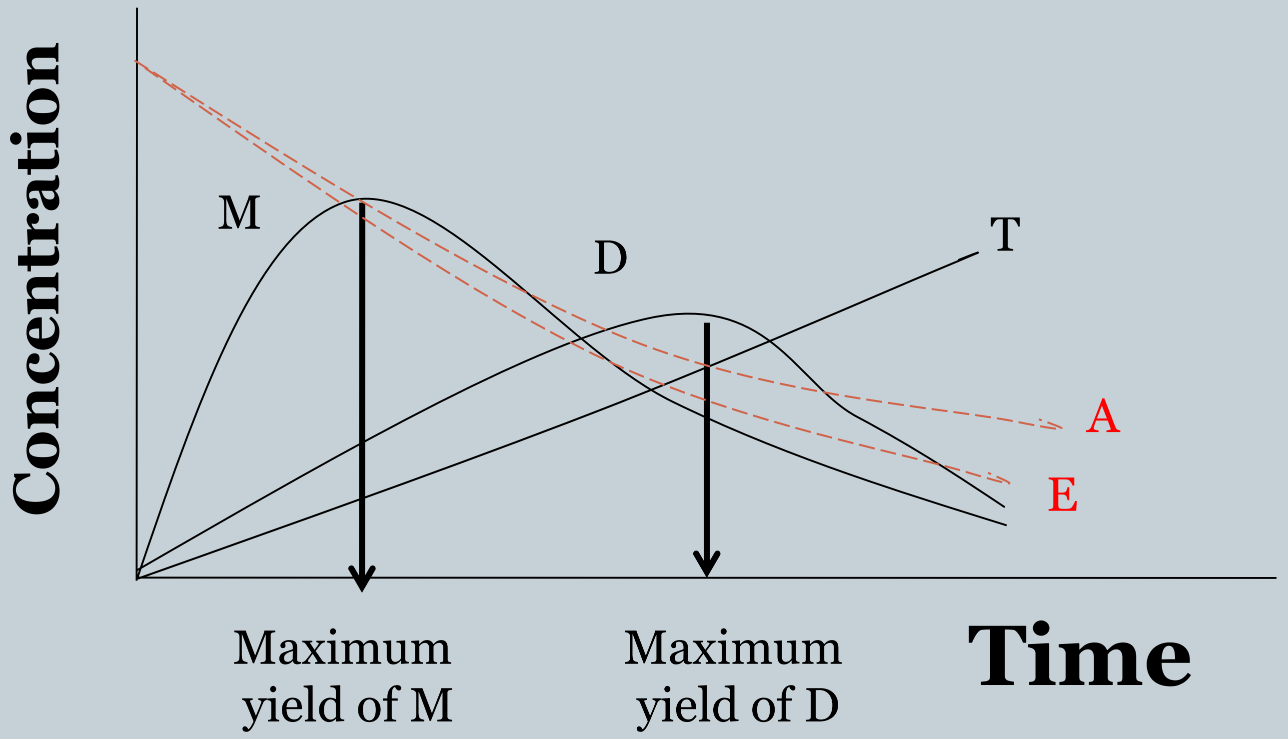
As T $\uparrow$, k $\uparrow$ As C $\uparrow$, k $\uparrow$

- On using **high ratio** of A : E the **yield of M** is **increased**.

- On using **lower ratio** of A : E the **yield of T** is **increased**.

Kinetics of Complex Series Reactions (cont.)

- **<u>Residence time</u> also affects the products**



Kinetics of Complex Series Reactions (cont.)

- **The average <u>residence time</u> for a <u>desired product</u> ratio may be picked up from <u>the graph</u>.**

- **If this gives a reactor with low throughput, change the temperature, pressure and/or reactant ratio.**

Typical Operating Conditions

Temp: 30 – 40 °C

Press: 10 – 20 psi

Mole ratio : Ammonia: ethylene oxide

10 : 1

Product distribution MEA 75 %

DEA 21 %

TEA 4 %

Process Description

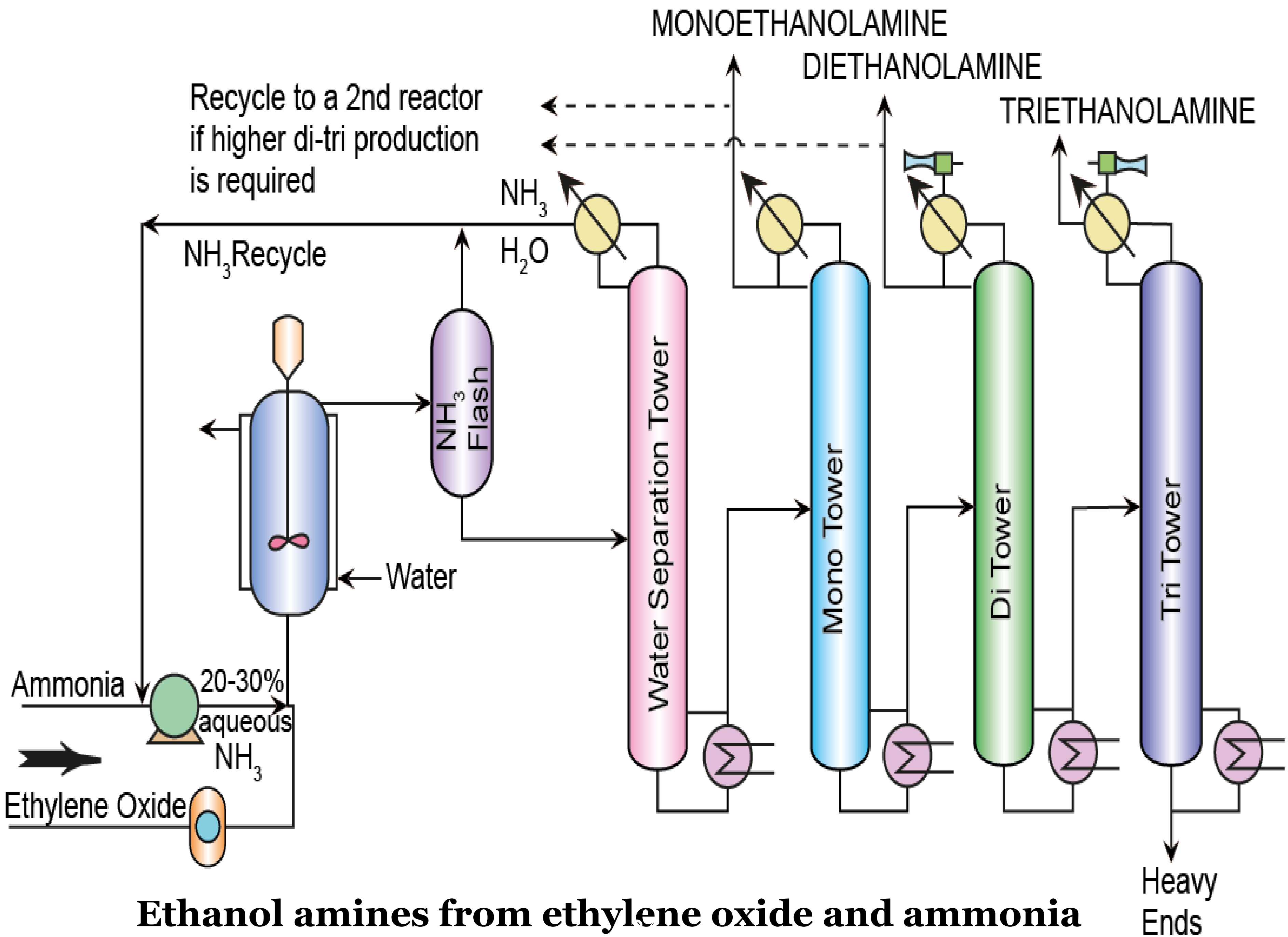
- Ethylene oxide is compressed, mixed with NH_3 (20 – 30%) and fed to a stirred tank reactor. The reactor can be operated under a wide range of conditions according to the required product ratio.

- Ammonolysis is exothermic so heat must be removed using a jacketed reactor.

- Excess NH_3 is recycled to the feed.

Process Description (cont.)

- Water is separated in H_2O separation tower and is also recycled to feed.
- The product, mono-, di- and tri- ethanol amines are separated in three successive fractionating columns.
- If higher di- and tri- production is required, the mono-derivative is recycled to another reactor to which no NH_3 is fed.



Ethanol amines from ethylene oxide and ammonia

Major Engineering Problems (cont.)

- **Kinetics of complex series reactions**
  - Previously discussed.

- **Reactor design**
  - Kinetic considerations facilitate computer optimization by determining the rate constant data.
  - Reactors used are broad range conditions reactors.

E : A	0.5 – 3.0
Temp.	30 – 275°C
Press.	1 – 100 atm.

Major Engineering Problems (cont.)

- **Process alternates**

When di and tri ethanol amines are mainly required, mono ethanol amine is recycled to a separate reactor where only ethylene oxide is added. This allows formation of amino-ethers. Addition of CO_2 suppresses this reaction.

- **Recovery and purification**

Di and tri compounds have high boiling points (270, 360^{o}C, respectively), accompanied by decomposition. For that, vacuum fractionating – system is required for their separation.